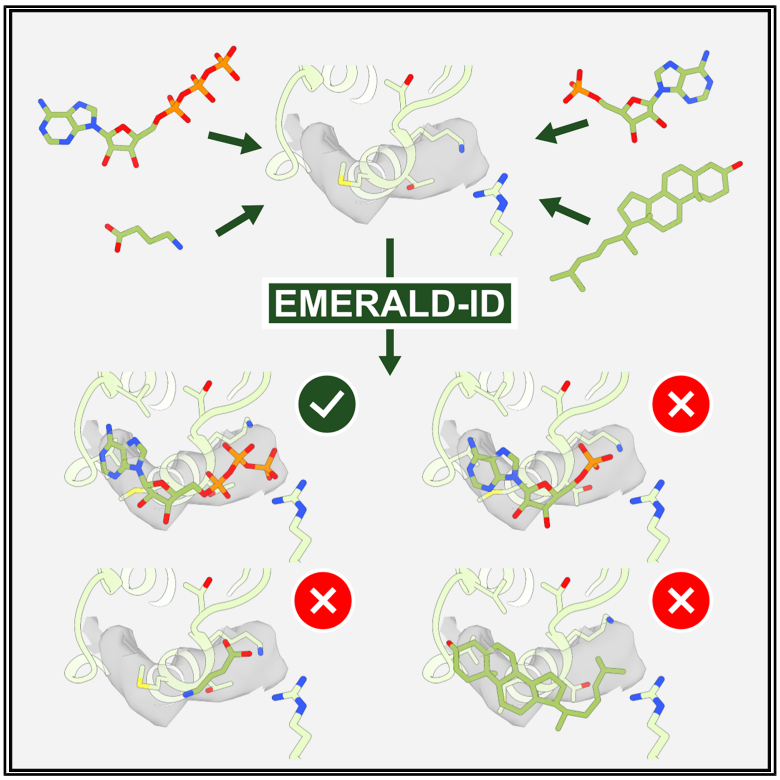
Resource

Structure

Automated identification of small molecules in cryoelectron microscopy data with density- and energy-guided evaluation

Graphical abstract



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In brief

Determining small molecule identities in cryo-EM data is vital for structural biology and drug discovery but is difficult to do manually. Muenks et al. developed EMERALD-ID, a method that selects an identity from a library of small molecules that is most likely to be present in the structure.

Highlights

- EMERALD-ID is a tool to determine ligand identity in cryo-EM maps
- EMERALD-ID combines a physical forcefield with density agreement to rank identities
- The method is benchmarked on deposited cryo-EM maps yielding 44% success
- EMERALD-ID finds several misidentified ligands in deposited structures

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Automated identification of small molecules in cryoelectron microscopy data with density- and energy-guided evaluation

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SUMMARY

Methodological improvements in cryoelectron microscopy (cryo-EM) have made it useful in ligand-bound structure determination for biology and drug design. However, determining ligand conformation and identity is challenging at the resolutions typical for cryo-EM. Automated methods can aid in ligand conformational modeling, but current ligand identification tools—developed for X-ray crystallography data—perform poorly at resolutions common for cryo-EM. Here, we present EMERALD-ID, a method capable of docking and evaluating small molecule conformations for ligand identification. EMERALD-ID identifies 44% of common ligands exactly and identifies closely related ligands in 66% of cases. We then use this tool to discover possible ligand identification errors, as well as previously unidentified ligands. Furthermore, we show EMERALD-ID identifying ligands from custom ligand libraries of various small molecule types, including human metabolites and drug fragments. Our method provides a valuable addition to cryo-EM modeling tools to improve small molecule model accuracy and quality.

INTRODUCTION

Over the past decade, cryoelectron microscopy (cryo-EM) has become widely used in macromolecular structure determination as advancements in both data collection and data processing have improved map resolutions. As EM data approaches atomic[1,2] and near-atomic resolutions, protein-small molecule interactions are observable, leading to an increase in ligands modeled in cryo-EM structures[3] and the use of cryo-EM in drug discovery.[4] Despite these advancements in data resolution, the lower resolution of typical cryo-EM maps means building models into cryo-EM data is still difficult and error-prone for both proteins[5] and ligands.[6]

Building on recent advances in protein structure prediction from machine learning, numerous tools exist for robustly building protein models into cryo-EM data.[7–12] However, tools for modeling ligands are less well-developed. While the cryo-EM ligand fitting tools GemSpot,[13] EMERALD,[14] and ChemEM[15] are available, no capable methods exist to accurately identify and fit ligands at moderate resolution data. Current automated ligand identification methods—primarily developed for crystallographic data—rely on density map correlations[16] or shape features of the maps,[17,18] leading to limited accuracy at resolutions worse than 3 Å. Deep learning has been successfully applied to ligand modeling problems with AlphaFold 3[19] and RoseTTAFold All-Atom,[20] and the MinkLoc3Dv2 deep learning architecture was utilized for ligand identification of cryo-EM maps.[21] However, AlphaFold 3 and RoseTTAFold All-Atom only determine ligand conformation, not identity, and are unaware of EM map information, while MinkLoc3Dv2 only provides ligand classes, with no ligand conformation generated.

In order to produce accurate small molecule models and determine ligand identity, we believe both map and energy features must be used. Here, we present EMERALD-ID, a ligand identification tool for cryo-EM data. EMERALD-ID utilizes the RosettaGenFF small molecule force field,[22] the EMERALD ligand fitting method,[14] and a linear regression model combining estimated binding affinity and density correlation to discern ligand identities from a user-provided library. The performance of EMERALD-ID was evaluated on ligand-bound structures of common ligands in the Electron Microscopy DataBank (EMDB),[23] upon which we found 55 EMDB entries where EMERALD-ID greatly favored an identity different from the deposited model. Additionally, we searched deposited maps in the EMDB and identified 108 maps with plausible ligand omissions. Lastly, we show the robustness of EMERALD-ID by screening against large, diverse libraries of human metabolites and drug fragments.

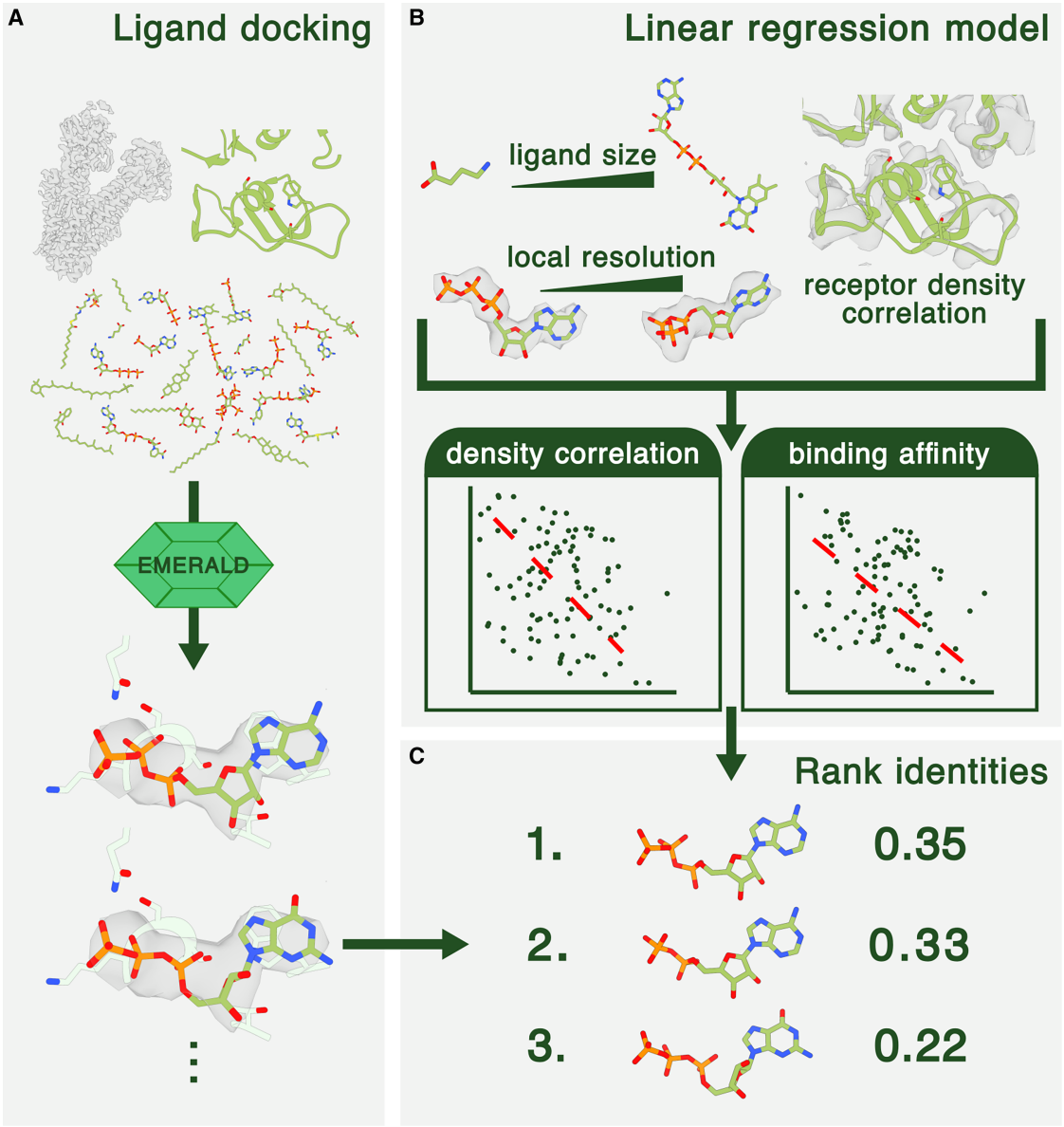


Figure 1. Overview of EMERALD-ID

(A) Identities from a user-provided library are fitted into the EM map with EMERALD.

(B) A linear regression model takes features of ligand size, local resolution, and receptor fit into density and predicts expected density correlation and binding affinity values for a given identity-map pair.

(C) Predicted values from the model are compared to calculated values from docked models to rank the identities.

RESULTS

Explanation of EMERALD-ID

An overview of EMERALD-ID is shown in Figure 1. EMERALD-ID takes a user-provided library of ligand identities, an EM density map, and a starting receptor model and docks all identities from the library into the EM map using EMERALD (Figure 1A). To compare identities fairly, we created a linear regression model that considers ligand size, local resolution of the map around the binding pocket, and the density correlation of the receptor to predict an expected ligand density correlation and estimated binding affinity (ΔG) for a given map and ligand (Figure 1B). The density correlations and Δ G values of all docked identities are compared to the expected values from the model to calculate unitless *Z* scores, which then determine a ranking of the molecules (Figure 1C).

To test EMERALD-ID, we wanted to focus on scenarios modelers may experience during structure determination. First, we created a ligand library of thirty common ligands in cryo-EM structures. With this library, we benchmarked EMERALD-ID against deposited cryo-EM structures, and we searched maps in the EMDB with unassigned density. Finally,

we examined EMERALD-ID's capabilities when considering a large endogenous ligand library, as well as its potential for fragment-based drug discovery (FBDD).

Evaluation of ligand identification in deposited structures

We first benchmarked EMERALD-ID's ability to identify ligands in density from a small pool of candidate ligands. While the popular modeling suite Phenix[16] provides a list of the most common ligands bound to macromolecular models, several of these ligands do not appear in any cryo-EM solved structures. To create a task relevant for cryo-EM, we curated our own list of common ligands found in cryo-EM structures. After filtering (see STAR Methods), we found 30 ligand identities that had 30 or more instances in the EMDB, which encompass 38% of small molecule structures in cryo-EM. This library included nucleotide substrates and cofactors such as ATP and NADH as well as lipids like cholesterol and palmitate. A full list of the molecules in the library is included in Table S1.

We gathered 1,386 appearances of a common ligand identity from 1,221 EMDB entries. All 30 ligands in the library were docked in the pocket of the first instance of the common ligand in the deposited structure. EMERALD-ID ranked the deposited identity first for 44% of instances (Figure 2A). Identification results were compared to phenix.ligand_identification that determined the deposited identity in 10% of cases (Figure 2A). In instances where EMERALD-ID did not identify the deposited ligand, it often chose a closely related identity. In 66% of entries, the top ligand had a Tanimoto similarity coefficient greater than 0.75 to the deposited identity. EMERALD-ID often confused nucleotides that differed by phosphate length or base, which are ambiguous at medium to low map resolutions (Figure 2B). For steroids and lipids, EMERALD-ID tends to favor smaller ligands within the class, e.g., cholesterol (CLR) vs. cholesterol hemisuccinate (Y01) (Figure 2B), which is expected given that larger ligands likely have disordered regions that are not represented in the EM map and have larger conformational spaces to explore when docking.

Our ability to identify the deposited ligand relied heavily on successfully docking the molecules. EMERALD-ID docked the native identity within 1 Å root-mean-square deviation (RMSD) of the deposited structure for 39% of cases; the deposited ligand ranked first in 69% of these cases, while a similar ligand ranked first in 90% of them (Figure 2C). If the deposited structure was included during EMERALD docking for the cases with an RMSD above 2 Å, the identification success did not improve, going from 23% to 21%. This suggests that the docking failures are not from a lack of sampling, but that our score function does not think the deposited ligands in these cases were the lowest energy conformation. The ability to rank the deposited ligand first decreased as the local resolution of the binding pocket decreased (Figure 2D). The deposited ligand was ranked first for 47% for all maps with 4.5 Å resolution or better, but that rate plummeted at worse resolutions. This was unsurprising, given the lack of detail and increased uncertainty in maps at low resolutions, and we previously showed that ligand fitting performance in EMERALD decreased at this same resolution.[14] The fraction of similar identities ranked first does not decrease as sharply with resolution, suggesting EMERALD-ID is robust at identifying a ligand class even at low resolutions.

When considering identification success, we also looked at the Z score of the top-ranked identity, and the Z score gap between the first- and second-ranked ligand. We found that identification success improved as the Z score of the top-ranked identity increased (Figure 2E) and as the gap between the first- and second-ranked ligand increased (Figure 2F). By examining identities with Z scores above -0.5 and a gap greater than 0.25 to second place, we found 55 cases where EMERALD-ID chose an identity different than the deposited model, but was confident in its prediction. Nineteen of these disagreements were between cholesterol and the larger steroid molecules 1N7, 9Z9, and Y01. Another common possible correction occurred between ATP and ADP, which occurred 8 times. For example, in an intermediate state of the MutS protein,[25] EMERALD-ID prefers an ATP molecule over the deposited ADP molecule. The deposited model does not fully explain the observed density (Figure 3A), while the EMERALD-ID conformation satisfies the density while forming an extensive hydrogen bond network (Figure 3B).

Another example of ambiguous nucleotide identity was found in a structure of the Ufd1/Npl4/Cdc48 complex.[26] In the deposited structure, the modeled ATP molecule satisfied the EM map, but in doing so placed the gamma phosphate near an aspartate residue (Figure 3C). EMERALD-ID elected to avoid this repulsive clash and left a portion of the map unexplained with the top-ranked ADP molecule (Figure 3D). Likely, the unexplained density belongs to a magnesium ion (Figure S1). Even if EMERALD-ID did not explicitly model the ion, it avoided overfitting into the density because the conformation does not fit energetically.

A final example includes a molecule that was too large for the observed density. In a malic enzyme 2 structure,[27] the nicotinamide moiety of the NAD^+ cofactor was unsupported by the density map in the deposited structure (Figure 3E). The binding pocket is a general nucleotide binding site,[27] so the AMP molecule favored by EMERALD-ID satisfied the nucleotide restriction while having a better fit into the EM map (Figure 3F).

Comparison of EMERALD-ID to the MinkLoc3Dv2 deep-learning model

With deep-learning methods becoming more prevalent in structure modeling, we wanted to compare EMERALD-ID's performance to them. We chose to compare EMERALD-ID to a ligand classifier model using the MinkLoc3Dv2 architecture[21] as cryo-EM data was included in its training. There are fundamental differences between EMERALD-ID and the MinkLoc3Dv2 model that make comparison difficult. The MinkLoc3Dv2 model only provides a ligand class label, while EMERALD-ID provides both an identity label and a conformation fitted to the density. Moreover, the MinkLoc3Dv2 model considers a fixed library of 219 ligand classes for identification, many of which cannot be docked in EMERALD such as ions and covalently bound glycans.

For comparison, then, we created a library of 103 ligand identities that can be considered in both methods, and tested both methods on a subset of 291 EMDB entries. EMERALD-ID ranked the deposited identity first for 36% of cases. The MinkLoc3Dv2 model predicted the ligand class of the deposited identity in

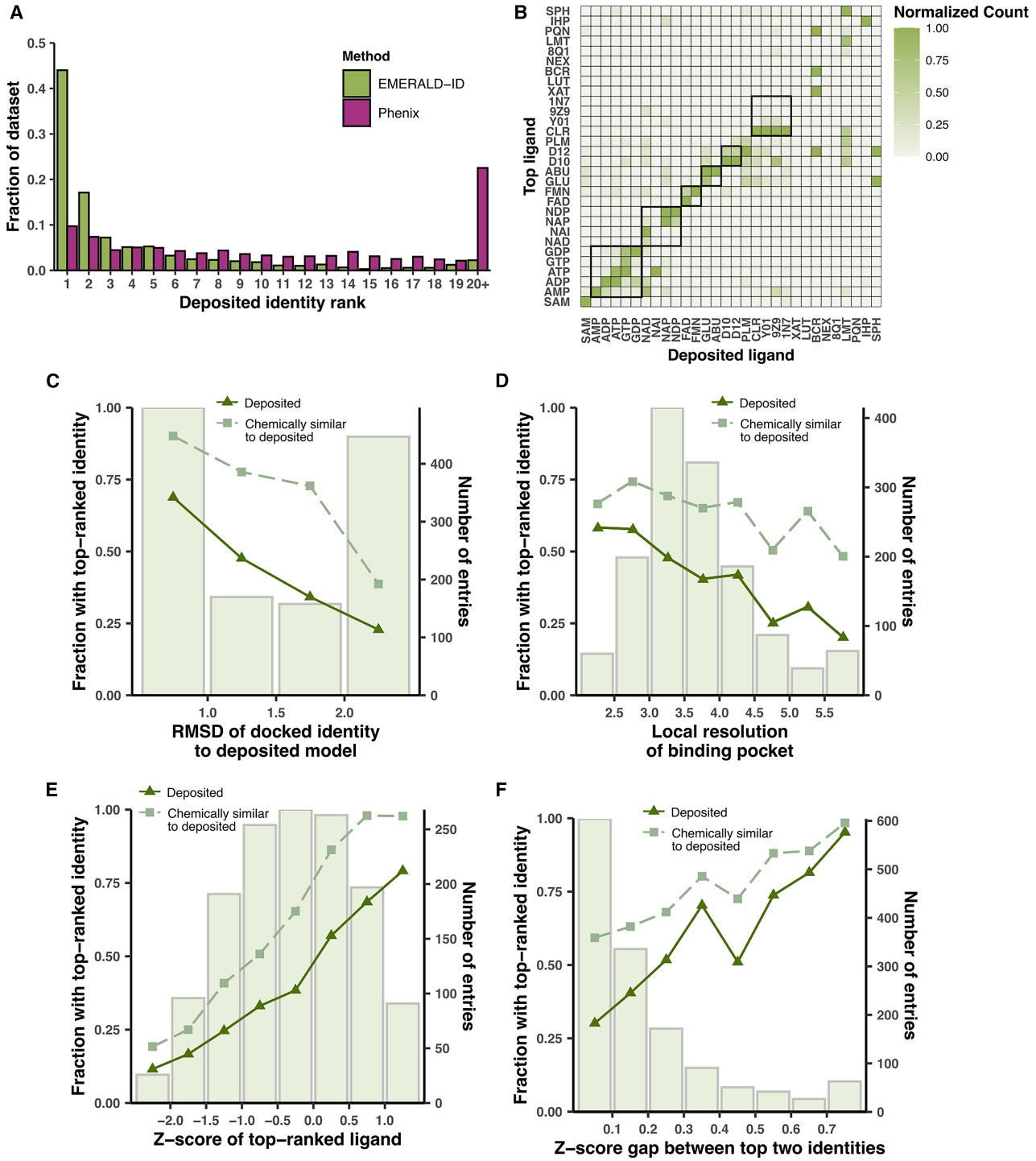


Figure 2. EMERALD-ID results from screening common ligand identities

(A) Fraction of the data for ranks of the deposited identity for EMERALD-ID (green, n = 1386) and Phenix (purple, n = 1030).

(B) Confusion matrix of common ligand identities. Ligands are labeled by their name in the Chemical Component Dictionary[24] and are ordered by Tanimoto similarity. Boxes are drawn around ligands belonging to a similar class or shape. Counts for each identity are normalized by column.

(C–F) Ability to rank the deposited ligand first (triangle, solid) or a similar identity first (Tanimoto score > 0.75; square, dashed) by docking success (C), local resolution of the binding pocket (D), Z score of the top-ranked identity (E), and the gap between the first- and second-ranked identities (F). Frequency of each bin are shown in the background. See also Figure S2 and Table S1.

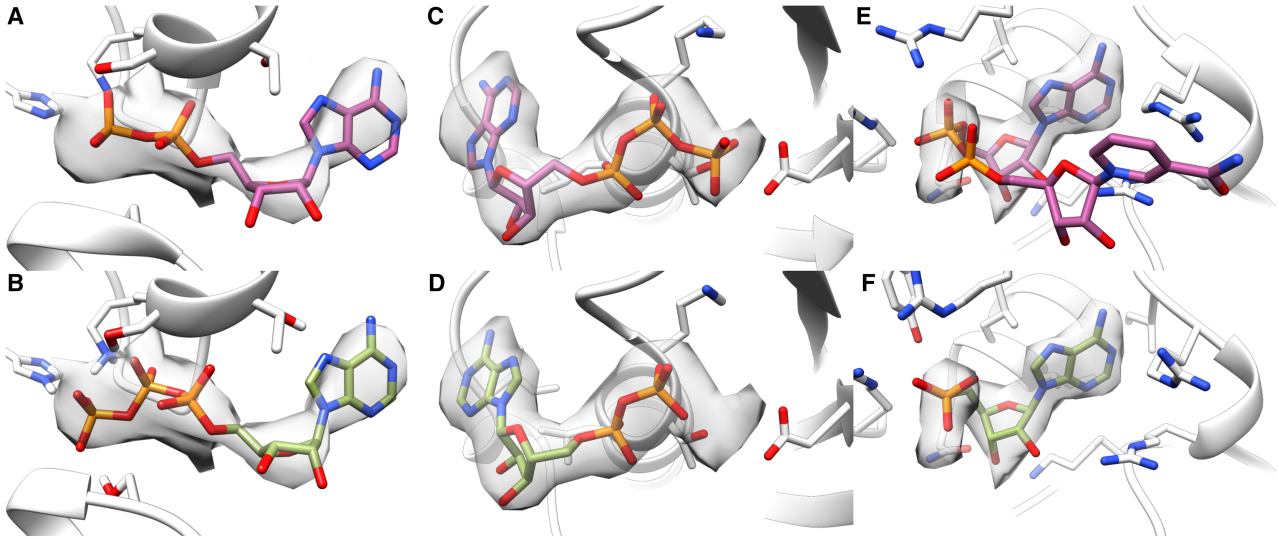


Figure 3. Examples of high-confidence EMERALD-ID identities different than the deposited model

(A and B) Deposited ADP molecule (A) is disfavored for an ATP molecule in EMERALD-ID (B) in MutS (EMDB: 11793, PDB: 7AI7, chain: A, residue: 1801).
(C and D) Deposited ATP molecule (C) is replaced with an ADP molecule in EMERALD-ID (D) in the Ufd1/Npl4/Cdc48 complex (EMDB: 27273, PDB: 8DAR, chain: A, residue: 901). Nearby aspartate residue shown on the right.
(E and F) Deposited NAD^+ molecule (E) in malic enzyme 2 (EMDB: 33145, PDB: 7XDE, chain: A, residue: 601) is outscored by an AMP molecule (F).
Map for (A and B) is zoned 2.5 Å around ligand at a 0.268 contour level ($\sigma = 1.5$). Map for (C and D) is zoned 2.0 Å around ligand at an 8.82 contour level ($\sigma = 0.5$). Map for (E and F) is zoned 3.0 Å around ligand at a 1.35 contour level ($\sigma = 0.8$). See also Figure S1.

27% of entries, but this increases to 49% when labels that cannot be modeled in EMERALD-ID were disregarded. The MinkLoc3Dv2 model outperforms EMERALD-ID in cases where ADP, ATP, and cholesterol were the deposited ligand, while both methods perform similarly for other identity classes (Figure S2). These three ligands were the most abundant in our common ligand search, so it is likely the MinkLoc3Dv2 model has seen more examples of density for these ligands, leading to better performance.

Discovery of unassigned density of deposited EM maps

Given the low resolution of cryo-EM maps and lack of ligand identification tools, we suspected that several deposited maps contained regions of density corresponding to ligands that were left unidentified. To remedy this, we searched the EMDB for small molecule-sized unmodeled map regions and screened them using the library of common ligands. Detected regions were filtered by their volume and proximity to the macromolecule so that only the most likely ligand regions were searched. We detected 296 regions from 108 map entries with a Z score above -0.5 for the top-ranked ligand. These entries were deemed most likely to contain a ligand and were manually inspected for identity assignment. Likely identifications are shown in Figure 4.

Nucleotide di- and triphosphates were commonly found as unmodeled ligands. In the CLC-7/Ostm1 antiporter,[28] the EM map shows nucleotide-like density, and EMERALD-ID produced an ADP model that fit the map and interacted with the nearby phosphate binding loop (Figure 4A). This evidence, the confidence of EMERALD-ID, and that ATP was modeled at this site in a higher resolution map[29] (Figure 4B) all supported this as a nucleotide binding site. We also identified an ATP molecule at an apparent nucleotide binding site in a known ATPase[30] (Figure 4C) that likely went unmodeled because the ligand pocket was not of interest for this protein structure. A structure from the same study modeled ATP at this binding pocket as well (Figure 4D). Lastly, we found density in a structure of an Na-K-Cl cotransporter in zebrafish[31] that EMERALD-ID suspected as an ADP molecule (Figure 4E). Since this structure's publication, a nucleotide binding site has been determined in the C-terminal domain in the human homolog of the cotransporter[32] (Figure 4F), supporting our finding.

Along with nucleotides, our unmodeled density detection found several possible lipid identifications. EMERALD-ID often suggested palmitate molecules in coronavirus spike proteins (Figure 4G). It is known that spike proteins have fatty acid binding sites[33] (Figure 4H), and we previously used EMERALD to model linoleic acid in a spike protein.[14] While it is likely that palmitate is not the exact identity, we detected the signal of a fatty acid binding site nonetheless.

In addition to palmitate molecules, we also found that ten and twelve carbon chain lipids often ranked highly in detected density. For two examples of transient receptor potential vanilloid (TRPV) channels,[34,35] the density was found in the transmembrane region of nanodisc-reconstituted proteins (Figure S3) and likely corresponded to disordered lipids from the nanodiscs that cannot be fully identified. While we cannot confidently assign an identity, the frequency of detected regions such as these and the abundance of membrane protein structures solved by cryo-EM suggest that lipids go undetected in EM maps.

Identifying uncommon ligands using an endogenous ligand library

While we detected several ligand identities with the common ligand library, we found other density regions that looked like

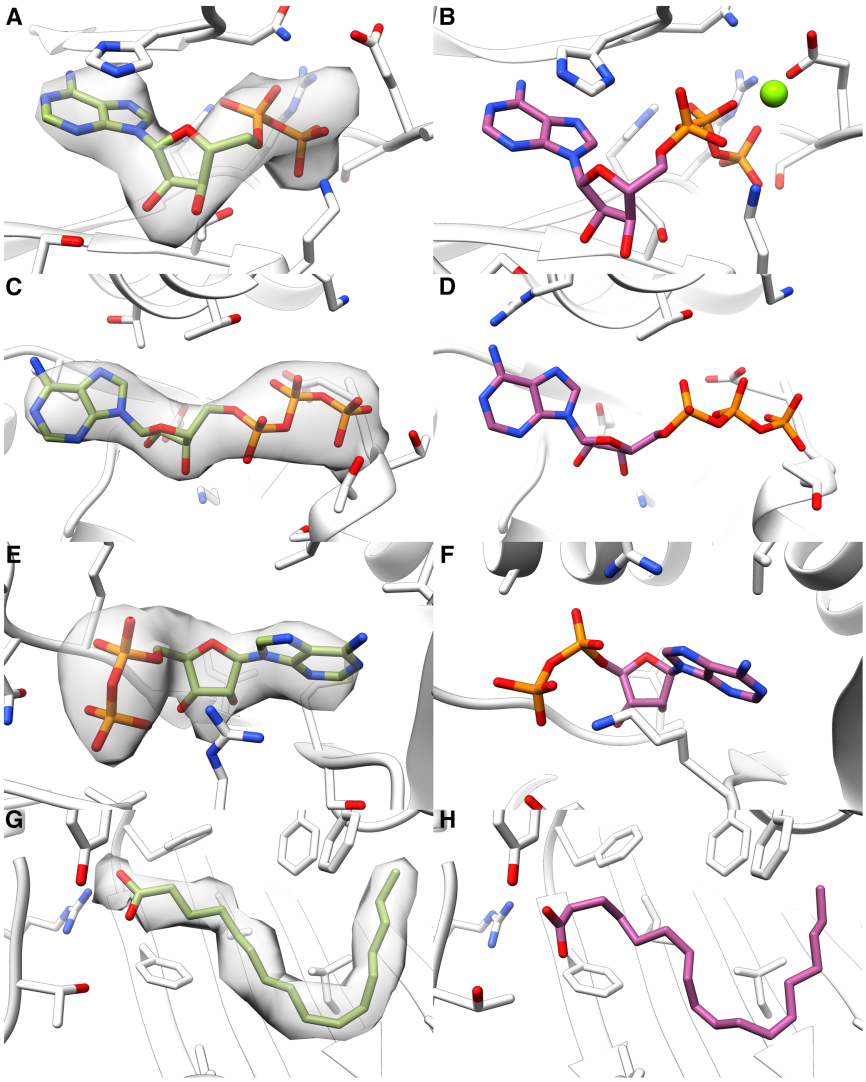


Figure 4. Likely ligands identified in unmodeled regions of deposited maps

(A) Detected ADP molecule in the CLC-7/Ostm1 antiporter (EMDB: 30238, PDB: 7BXU).

(B) ATP molecule bound at the same site in (A) in a higher resolution EM structure (PDB: 7JM7).

(C) Found ATP molecule in the TRiC complex (EMDB: 33053, PDB: 7X7Y).

(D) ATP molecule at an identical site of (C) in a TRiC complex structure from the same study (PDB: 7X3J).

(E) ADP molecule ranked first in detected density for a zebrafish Na-K-Cl cotransporter (EMDB: 0473, PDB: 6NPL).

(F) ADP bound at nucleotide binding site of a human Na-K-Cl cotransporter (PDB: 7AIQ).

(G) Detected palmitate molecule bound to a spike protein of SARS-CoV-2 (EMDB: 11207, PDB: 6ZGI).[54]

(H) Linoleic acid bound in the free fatty acid binding pocket in SARS-CoV-2 spike protein (PDB: 6ZB5).

Map for (A) is zoned 3.0 Å around ligand at a 0.031 contour level (σ = 0.85). Map for (C) is zoned 3.0 Å around ligand at a 0.418 contour level (σ = 1.0). Map for (E) is zoned 3.0 Å around ligand at a 0.0664 contour level (σ = 1.2). Map for (G) is zoned 2.5 Å around ligand at a 0.744 contour level (σ = 0.6). See also Figure S3.

ligands, but evaluation with common ligand identities provided inadequate models. Additionally, microscopists may co-purify an unknown endogenous ligand with a protein sample, which requires a larger ligand library for identification. To cover scenarios that require more ligand identities, we increased the size of the provided library from 30 to 2,950 molecules and tested EMERALD-ID's capability on 7 cryo-EM structures containing an uncommon ligand.

To determine test cases, we searched the EMDB for entries containing one of the 2,950 detected metabolites from the Human Metabolome Database (HMDB)[36] and looked at rare ligands with 3 or fewer instances in EM structures. After further filtering by ligand size, resolution, and specimen species, we found 14 cases containing an uncommon ligand, which were reduced to 7 after manual inspection. A summary of the EMERALD-ID results for these cases are shown in Table 1, with additional information such as 2D drawings of molecules included in Figure S4. EMERALD-ID ranked the deposited ligand in the top 10% in three out of seven cases (Table 1). For a fourth case (EMDB: 14725, PDB: 7ZH6), the top 10 identities all shared the same steroid core as the endogenous ligand model. Of these four cases, all contained a ligand in the top 5% with a Tanimoto similarity coefficient above 0.75, with three being in the top 1%. There were three cases with low signal for the deposited identity. For one (EMDB: 34910, PDB: 8HNC), the binding pocket contained density regions unexplained by the deposited ligand (Figure S4C). The unexplained density allowed flexible ligands, such as the top-ranked phosphatidylinositol molecule (Figure S4A), to fit into the map and have high-ranking density *Z* scores. The other two failures (EMDBs: 38692 and 38966; PDBs: 8XV5 and 8Y65) were small ligands with low density signal in the EM map (Figures S5D and S5E). This led to low *Z* scores for all molecules in the library (Figure S4B), with no density *Z* score being above −0.5.

During our search for unmodeled ligands in the previous section, we found instances in the EM map that appeared ligand-like, but none of the common ligands scored well. We decided to screen these regions with the endogenous ligand library to find more probable identity matches. For a Piezo 1 ion channel,[37] a sphingosine lipid was ranked first using the common ligands (Figure S5A). While the ligand is likely a lipid, the sphingosine model left unexplained density, and the density shape and nearby arginine residues suggested a phospholipid identity. Following identification with the HMDB library, the top-ranked molecule was a phosphatidylserine lipid that explained the binding pocket well (Figure 5A). EMERALD-ID detected a glutamate ligand for an ADH3 structure in *Stenotrophomonas*

Table 1. Endogenous library screen of uncommon ligand identities

Entry	Rank	Rank of similar ligand
EMD-30074[55]: HC3	7/2701 (0.3%)	1/2701 (0.04%)
EMD-25195[56]: BUA	33/2564 (1.3%)	15/2564 (0.6%)
EMD-14725[57]: C0R	314/2676 (11.7%)	15/2676 (0.6%)
EMD-34910[58]: BLR	635/2753 (23.1%)	635/2753 (23.1%)
EMD-35017[59]: GCO	170/2804 (6.1%)	93/2804 (3.3%)
EMD-38692[60]: PXM	608/2516 (24.2%)	608/2516 (24.2%)
EMD-38966[61]: URC	1364/2593 (52.6%)	1364/2593 (52.6%)

Similar ligand defined as an identity with a Tanimoto similarity coefficient greater than 0.75 to the deposited model. See also Figure S4.

acidaminiphila[38] (Figure S5B). Another structure of the protein in the same study contains a phenylalanine at this binding site,[38] which was not included in the common ligand library. However, EMERALD-ID detected the amino acid signal, and when a larger library was included, ranked several phenylalanine derivatives within the top 10 structures (Figure 5B).

We also detected a conspicuous ligand blob at the benzodiazepine binding site in a γ-aminobutyric acid ($GABA_A$) receptor[39] (Figure S5C). Drugs in the benzodiazepine class bind extracellularly to $GABA_A$ receptors causing sedative effects, making benzodiazepines effective drugs for anesthetics, seizures, and psychiatric conditions.[40–42] Given the site's pharmacological importance, endogenous ligands for the site have been sought after, with no known small molecules acting as functional endogenous binders. When we performed an endogenous ligand screen on the detected region, we found 2 plausible identities. Inosine ranked third overall (Figure 5C). Inosine was found to bind to the benzodiazepine site of $GABA_A$ receptors,[43,44] but has been discredited as an endogenous binder for weak binding and lack of activity.[45] We also found the neurosteroid allopregnanolone in the top 2% (Figure 5D). Allopregnanolone was included in the sample preparation of the structure and appeared in the transmembrane region of the deposited model,[39] where it is known to modulate $GABA_A$ receptor activity.[46,47] While further experiments will be needed to confirm the ligand identity, EMERALD-ID provided two reasonable explanations of a small molecule bound to the benzodiazepine site.

Identifying fragments for drug screening experiments

EMERALD-ID proved effective when distinguishing identities of endogenous ligands, but as cryo-EM becomes more relevant for drug discovery,[48] generalizing the method for drug identification becomes crucial. Ligand identification is a necessary task during FBDD. In FBDD, low molecular weight molecules that weakly bind to a drug target are determined and used as a scaffold to build a drug candidate. An important step in FBDD is to obtain a structure of a fragment bound to the target. However, the identity of the bound fragment may be unknown if a cocktail of fragments is included during sample preparation. Traditionally, structure determination for FBDD has been achieved through X-ray crystallography because high-resolution is needed to resolve the identity of the ligand. But many drug targets contain transmembrane regions, precluding the use of X-ray crystallography for their structure determination. As resolution limits improve in cryo-EM, it is possible to obtain EM maps with resolvable fragment density—opening FBDD to drug targets that are difficult to solve with X-ray crystallography.

Principles of FBDD were successfully applied to determine high-resolution fragment bound structures by Saur et al.[49] They resolved 4 structures of fragment-sized ligands bound the cancer target PKM2, two of which included cocktails of 4 fragments during sample preparation. These two structures provided examples for us to test fragment screening with EMERALD-ID. We included both cocktails as libraries for their respective EM maps. EMERALD-ID identified both of the fragments determined by the original authors (Table 2).

While these results are promising, more examples will be needed to evaluate EMERALD-ID's utility for FBDD. To provide more test cases at lower resolution, we turned to realistic simulation EM data (details in STAR Methods). We found 7 high-resolution structures solved by X-ray crystallography that contained fragment-sized ligands and simulated EM data for them at 3.5 Å resolution. The native fragments were screened against a combined 238 fragment library from the Cambridge[50] and York[51] 3D libraries. For five of the seven entries, EMERALD-ID ranked the native fragment within the top 5 structures (Table 2). Moreover, fragments ranked above the native fragments share similar characteristics to the native fragment (Figures S6H and S6I). Despite binding weakly to their receptors, the binding affinity Z score was powerful at discerning between identities, with 5/7 native fragments ranking first by this metric (Table 2). This suggests that EMERALD-ID can be used for fragment screening, even when the fragment density is poor.

DISCUSSION

Here, we introduce EMERALD-ID to assign identities to ligand density in cryo-EM data. We identified the deposited ligand in over 40% of instances that contained a common ligand, a rate much higher than Phenix ligand identification. The power of EMERALD-ID was further shown by identifying several ligands that were left unmodeled during the original deposition. Finally, EMERALD-ID proved effective in plausible scenarios of screening a large endogenous ligand library and a fragment library for drug discovery.

Given how it corresponded with identification success, we believe the values of the Z scores will be useful when evaluating ligands. We find that 77% of entries in the common ligand benchmarking set have a top scoring identity with a Z score of -1.0 or greater. Meanwhile, only 21% of our detected unmodeled ligand regions found a ligand identity better than this threshold. This suggests that the Z score is sensitive to whether a ligand is present in the structure. Additionally, both binding affinity and density Z scores should be above -1.0 to eliminate ligands that overfit to the map or ignore the map. The binding affinity calculations have a slight bias toward large hydrophobic ligands, and molecules with 10 or fewer heavy atoms can benefit from high density correlations from overfitting. We recommend caution if either scenario describes the top identity and suggest using the Z score guidelines described previously to interpret results.

As noted, the success of identification with EMERALD-ID greatly depended on the success of docking the small molecules

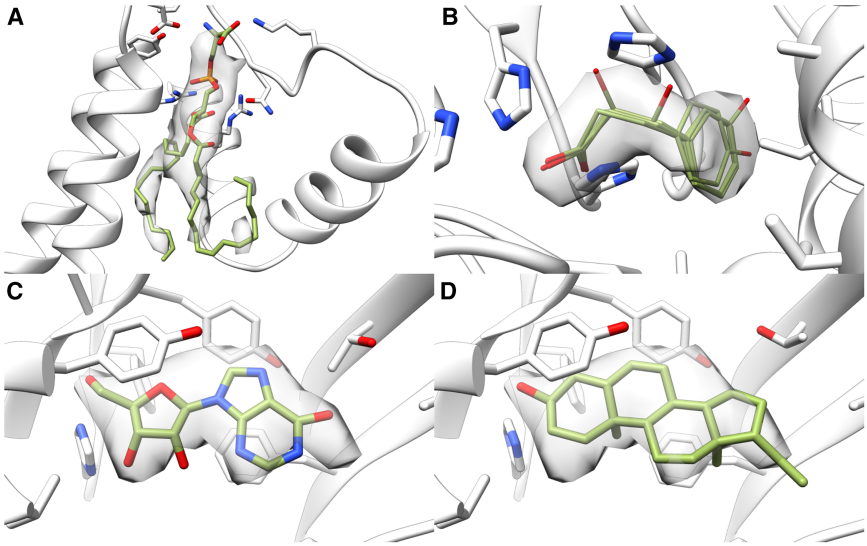


Figure 5. Endogenous ligand search of detected density

(A) Top-ranking phosphatidylserine molecule for detected density in the Piezo 1 ion channel (EMDB: 7128, PDB: 6BPZ).

(B) Identities ranked in the top 10 that share features to phenylalanine in ADH3 from *S. acidaminiphila* (EMDB: 35452, PDB: 8IHQ).

(C and D) Well-scored identities for detected density at the benzodiazepine binding site in a $GABA_A$ receptor (EMDB: 40462, PDB: 8SGO). Inosine (C) ranked third overall and allopregnanolone (D) ranked 48th.

Map for (A) is zoned 2.0 Å around ligand at a 0.0202 contour level (σ = 0.35). Map for (B) is zoned 2.5 Å around ligand at a 0.26 contour level (σ = 0.6). Map for (C and D) is zoned 2.0 Å around ligand at a 0.0369 contour level (σ = 1.1). See also Figure S5.

into density, and the limitations of the method mainly lie with limitations in our ligand docking. Molecules containing inorganic elements such as metal ions cannot be properly parameterized for our ligand docking, leaving out ligands such as hemes from analysis. Another exclusion from analysis are glycans which, due to their unique structural characteristics and covalent binding, require special methods for docking and identification. Additionally, EMERALD can only dock a single ligand identity at once, so pockets with coexisting identities or cofactors must be docked successively.

Misidentification may still occur even if the true identity is well fitted. However, most identity confusion in EMERALD-ID occurred between similar identities (Figure 2B; Table 1), so even if the true identity is not ranked first or included in the library, a ligand in the same class will likely score well. Moreover, there may be an insignificant gap in *Z* score between the top ligand and other identities. We found this gap was less than 0.1 *Z* score for 44% of our benchmarked cases. If the *Z* scores among multiple identities are close to one another, it may be appropriate to consider multiple identities or investigate the identities with other experiments.

Improvements to EMERALD-ID will likely come from changes in the force field in Rosetta due to the method's reliance on binding affinity calculations. As mentioned previously, these calculations prefer flexible lipids. Corrections to hydrophobic interactions in Rosetta or other advancements in binding affinity calculations via deep learning will alleviate these issues. While our simple linear regression model is effective in estimating binding affinity and ligand map correlation, the model will likely improve with better training data and additions of predictive features—which should occur with standardization of EM tools for ligand validation.[3,52,53] Additionally, the model could be optimized for specific identification tasks or libraries. As presented, EMERALD-ID is effective in identity determination for general modeling scenarios, and we hope that its ligand placements and *Z* score calculations contribute to improved quality of ligand models for better insights into structural biology.

Table 2. Screening drug fragments for real and simulated EM data

Saur et al. Entry (EMDB ID)	Rank	ΔG *Z* score rank
EMD-10577: NXE	1/4	3/4
EMD-10584: NXH	1/4	1/4
Simulated data Entry (PDB ID)	**Rank**	**ΔG *Z* score rank**
1EQG [62]: IBP	1/236	1/236
1GWQ [63]: ZTW	1/237	1/237
1N1M [64]: A3M	2/237	1/237
1QWC [65]: 14W	3/237	25/237
1S39 [66]: AQO	1/232	1/232
1YZ3 [67]: SKA	36/236	1/236
2OHK [68]: 1SQ	20/236	42/236

See also Figure S6.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Frank DiMaio (dimaio@uw.edu).

Materials availability

This study did not generate new materials.

Data and code availability

- Data used for this study are publicly available as of the date of publication at https://doi.org/10.5281/zenodo.15122660, organized by their respective figure or table. These data include source data for each plot and underlying data used to generate them, all docked ligand conformations for structures featured in figures, docked ligand conformations of the deposited and top identities for the common ligand screen, and docked structures for every tested identity for the endogenous ligand and drug fragment screens. Projection stacks, metadata files, and reconstructed maps for simulated EM data for Table 2 are available for download at https://files.ipd.uw.edu/pub/EMERALD-ID/Table~2.tar.gz, and the same files for Figure S6 are available at https://files.ipd.uw.edu/pub/EMERALD-ID/Table~2FigS3.tar.gz.

- Code for EMERALD-ID, the unmodeled density detector, and the cryo-EM density simulation are all available in Rosetta for weekly releases after November 12, 2024. Instructions on how to use them and example scripts used for this manuscript are publicly available as of the date of publication in the tutorials file at https://doi.org/10.5281/zenodo.15122660.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

A.M., conceptualization, methodology, software, validation, formal analysis, investigation, writing – original draft, writing – review & editing, and visualization. D.P.F., software and writing – review & editing. G. Z., software and writing – review & editing. F. D., conceptualization, methodology, software, writing – original draft, writing – review & editing, and funding acquisition.

DECLARATION OF INTERESTS

D.P.F. is currently employed at Johnson & Johnson.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
MutS intermediate state	Fernandez-Leiro et al., 2021[25]	PDB: 7AI7 EMD: 11793
Ufd1/Npl4/Cdc48 complex	Lee et al., 2023[6,26]	PDB: 8DAR EMD: 27273
Malic enzyme 2	Hsieh et al. 2023[27]	PDB: 7XDE EMD: 33145
CLC-7/Ostm1 antiporter	Zhang et al. 2020[28]	PDB: 7BXU EMD: 30238
CLC-7/Ostm1 antiporter	Schrecker et al. 2020[29]	PDB: 7JM7
TRiC complex	Liu et al. 2023[30]	PDB: 7X7Y EMD: 33053
TRiC complex	Liu et al. 2023[30]	PDB: 7X3J
zebrafish NKCC1 cotransporter	Chew et al. 2019[31]	PDB: 6NPL EMD: 0473
human KCC1 cotransporter	Chi et al. 2021[32]	PDB: 7AIQ
SARS-CoV-2 spike protein	Wrobel et al. 2020[54]	PDB: 6ZGI EMD: 11207
SARS-CoV-2 spike protein	Toelzer et al. 2020[33]	PDB: 6ZB5
Piezo 1	Saotome et al. 2018[37]	PDB: 6BPZ EMD: 7128
ADH3	Dai et al. 2023[38]	PDB: 8IHQ EMD: 35452
$GABA_A$ receptor	Legesse et al. 2023[39]	PDB: 8SGO EMD: 40462
Scap/Insig complex	Yan et al. 2021[55]	PDB: 6M49 EMD: 30074
SMCT1	Han et al. 2022[56]	PDB: 7SL9 EMD: 25195
OCT3	Khanppnavar et al. 2022[57]	PDB: 7ZH6 EMD: 14725
OATP1B1	Shan et al. 2023[58]	PDB: 8HNC EMD: 34910
TMEM87A	Kang et al. 2024[59]	PDB: 8HTT EMD: 35017
SLC19A3	Dang et al. 2024[60]	PDB: 8XV5 EMD: 38692
GLUT9	Shen et al. 2024[61]	PDB: 8Y65 EMD: 38966
PKM2 with compound 6	Saur et al. 2020[49]	PDB: 6TTI EMD: 10577
PKM2 with compound 10	Saur et al. 2020[49]	PDB: 6TTQ EMD: 10584
Ovine COX-1	Selinksy et al. 2001	PDB: 1EQG
Estrogen receptor alpha	Warnmark et al. 2002	PDB: 1GWQ
Human Dipeptidyl Peptidase IV/CD26	Rasmussen et al. 2003[64]	PDB: 1N1M
Nitric-oxide synthase	Fedorov et al. 2003[65]	PDB: 1QWC
tRNA-guanine transglycosylase	Meyer et al. 2004[66]	PDB: 1S39
Human phenylethanolamine N-methyltransferase	Wu et al. 2005[67]	PDB: 1YZ3

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Beta-secretase	Murray et al. 2007[68]	PDB: 2OHK
TRPV2	Su et al. 2023[34]	PDB: 7YEP EMD: 33774
TRPV1	Zhang et al. 2021[35]	PDB: 7L2H EMD: 23128
Human Metabolome Database	Wishart et al. 2022[36]	https://hmdb.ca/
York 3D Fragment Library	Downes et al. 2020[51]	https://www.diamond.ac.uk/Instruments/Mx/Fragment-Screening/Fragment-Libraries/York-3D-Library.html
Cambridge 3D Fragment Library	Kidd et al. 2020[50]	https://www.diamond.ac.uk/Instruments/Mx/Fragment-Screening/Fragment-Libraries/Cambridge-3D-Library.html
Software and algorithms		
EMERALD-ID	This Study	https://rosettacommons.org/software/
sim_cryo	This Study	https://rosettacommons.org/software/
Unmodeled density detector	This Study	https://rosettacommons.org/software/
Open Babel v.3.1.0	O'Boyle et al. 2011[70]	https://openbabel.org/
Dimorphite-DL v.1.2.4	Ropp et al. 2019[71]	https://github.com/durrantlab/dimorphite_dl
MonoRes/Xmipp v.3.22.07.0	Vilas et al. 2018[72]	https://xmipp.i2pc.es/
Phenix v.1.21.1-5286	Terwilliger et al. 2007[16]	https://phenix-online.org/documentation/reference/ligand_identification.html
cryoSPARC v.4.4	Punjani et al. 2017[79]	https://cryosparc.com/
UCSF Chimera v.1.17.3	Goddard et al. 2007[80]	https://www.cgl.ucsf.edu/chimera/
ggplot2 v.3.4.3	Posit	https://ggplot2.tidyverse.org/
MinkLoc3Dv2 ligand classification	Karolczak et al. 2025	https://github.com/jkarolczak/ligand-classification
Figure and table source data and tutorials	This study	https://doi.org/10.5281/zenodo.15122660
Simulated EM data for Table 2	This study	https://files.ipd.uw.edu/pub/EMERALD-ID/Table2.tar.gz
Simulated EM data for Figure S6	This study	https://files.ipd.uw.edu/pub/EMERALD-ID/Table2FigS3.tar.gz

METHOD DETAILS

Ligand parameter generation

The outcome of ligand docking and identification greatly depends on the protonation state and partial charges assigned to the ligand. We recommend using our provided ligand parameters or using the MMFF94 force field[69] to calculate partial charges. For all experiments, hydrogen atoms were added at pH 7.4 to unprotonated SDF files via openbabel (v. 3.1.0).[70] Partial charges for the protonated files were calculated with the MMFF94 force field in openbabel. The resulting MOL2 files were then converted to Rosetta-specific ligand residue parameters files for docking. The origin of the unprotonated SDF file depended on the experiment. For model training, an SDF file of the first instance of the ligand in its respective structure was downloaded from the PDB. For the common ligand library, SDF files of the ideal geometries for each ligand were used.

The endogenous ligand library and drug fragment libraries started from SMILES strings. The SMILES for the endogenous ligand library were downloaded from the HMDB and converted into 2D coordinates with openbabel. The 2D coordinates were converted to 3D in openbabel with 3 successive rounds of 3D conformer generation on the slowest setting using a final energy minimization with 2000 steps of the steepest descent algorithm. Drug fragments had their SMILES strings protonated with dimorphite (v. 1.2.4)[71] at pH 7.4. The protonated SMILES strings were converted to 3D coordinates with openbabel on its default speed and then minimized with 2000 steps of the steepest descent algorithm.

Local resolution calculation

One feature used in training and evaluation of EMERALD-ID was the local resolution of the binding pocket. To generate these values for all maps across all experiments, we first calculated local resolution maps with MonoRes from the Xmipp software package (v. 3.22.07.0)[72] and then calculated average local resolution values for all voxels within 5 Å of the ligand's center of mass. Local resolution maps were determined by filtering the deposited EM map with a Gaussian kernel with a sigma of 0.02 times the map

dimensions. Voxels in the filtered map with a value above 0.05 times the maximum voxel value were saved to a binary mask for the map, which was then used by MonoRes to create the local resolution map. Voxels with local resolution values of zero were excluded from the average calculation. If the average local resolution in the binding pocket was more than 1 Å lower than the global resolution, then the global resolution was used in place of the local resolution.

Linear regression model calculation

To train the linear regression model, we took first instances of ligand identities in EMDB entries where the docked ligand conformation was within 1 Å RMSD of the deposited model in our previous EMERALD manuscript.[14] Since we only looked at ligands with 25 or fewer torsion angles when evaluating EMERALD, we supplemented the training data with entries that had ligand identities with over 25 torsion angles and could be docked within 1.5 Å RMSD of the deposited model. The ligands and surrounding flexible residues of all structures were relaxed in the EM map with a Cartesian minimization in Rosetta and their binding affinities and ligand map correlations were calculated. Ligand map correlations were transformed with an inverse hyperbolic tangent function so values lie within an infinite range. The relationships between these terms and ligand-map features were probed with a linear regression model in R (v. 4.3.1). A full summary of the linear regressions can be found in Tables S2 and S3, and diagnostic plots for the model are seen in Figures S7A–S7H.

We found that the number of heavy atoms in the ligand (a) predicted binding affinity with Equation 1

$$E(\Delta G) = -12.4442 - 0.4918a \quad \text{(Equation 1)}$$

and ligand density correlation could be predicted with the ligand's heavy atom count, the local resolution of the map 5 Å around the ligand, and the correlation of the entire receptor to the map with Equation 2

$$E(\operatorname{arctanh}(CC)) = 0.3428 - 0.03724r + 1.193p - 0.001455a \quad \text{(Equation 2)}$$

where r is the local resolution, p is the map correlation of the entire pose, and a is the number of heavy atoms. Scatter plots of each of these relationship are shown in Figures S7I–S7L.

The density correlations and binding affinities for all docked identities along with their respective expected values were used to calculate Z-scores where the expected value is the mean and the standard deviation was determined empirically by tuning the standard deviation of the residuals from the linear regression model ($\sigma_{\Delta G} = 12.64$, $\sigma_{density} = 0.1684$). The binding affinity Z-scores were negated so more positive Z-scores correspond to better binding affinities. Once Z-scores for the binding affinities and density correlations were calculated, they were combined into a composite Z-score.[73] There existed a correlation of 0.261 between the two scores, and we provide equal weight to both scores. Thus, the composite score was found by averaging the two values and dividing by $\sqrt{0.5+0.5(0.261)}$.

Determination of common ligand library

We wanted to provide a library of common ligands that can be used for most identification tasks. Common ligand libraries exist in other identification methods,[16,17] but these libraries were created from the entire PDB. Several ligands in the library are ligands relevant for X-ray crystallography but not cryoEM, like cryoprotectants. We decided to create our own library of common ligands specific for cryoEM solved structures. Entries from the EMDB between 2-6 Å global resolution for which a deposited ligand-bound model existed before September 13, 2023 were collected. The first instance of a unique ligand identity in each entry was counted, excluding ligands that cannot be processed by EMERALD, like covalently-bound ligands, ligands containing metal elements, and inorganic compounds. The resulting list of ligands contained several phospholipids. Identities among phospholipids are difficult to parse and require special considerations to dock properly due to their conformational search space, so we excluded examples of phospholipids from the common ligand library. Finally, analogs of higher count common ligands, such as ATP analogs phosphomethyl- and phosphoamino-phosphonic acid adenylate ester, were removed. The remaining ligands with more than 30 instances were separated and provided a library of 30 common ligand identities.

We searched the PDB for EM-solved entries that contain one of the 30 common ligands. Entries were filtered to exclude those with covalently-bound ligands, metal-coordinating bonds, and examples with another small molecule within 10 Å of the ligand of interest. Structures were further excluded if they were missing whole domains modeled into the map. After filtering, we had 1386 entries to screen common ligands with EMERALD-ID.

Small molecule docking with EMERALD

The EM map, ligand parameter files, and an input model of the receptor were provided to EMERALD for small molecule docking. Input models had all HETATM lines removed except for an ATP model centered on the analyzed ligand blob. The identity provided in the input structure does not matter, as long as the ligand is centered on the density that is being investigated. For each identity in the library, a pool of 100 ligand conformations were generated and optimized over 10 generations of a genetic algorithm as described in the EMERALD manuscript,[14] except when docking drug fragments where a pool size of 50 conformations were used because of their smaller conformational search space. The conformation with the lowest Rosetta energy for each identity was passed to EMERALD-ID for evaluation.

Estimated binding affinity values were calculated using a simple entropy model in Rosetta as described in Zhou et al.[74] For ligand-map correlation values, we applied a penalty to the value from the EMERALD-docked model because large ligands at low resolutions

had ligand map correlations unreasonably high for their fit into the map because of high background density signal from the receptor. The penalty was determined by the difference in map correlation with and without the ligand present (Δ lig_dens). The penalty was empirically derived by observing cutoffs of Δ lig_dens values from the training dataset and was calculated with Equation 3.

$$penalty = \begin{cases} 0 & \text{if } \Delta lig_dens \geq 0.15 \\ 0.15 - (2/3)\Delta lig_dens & \text{if } \Delta lig_dens < 0.15 \end{cases} \quad \text{(Equation 3)}$$

Once calculated, Z-scores were determined as explained above.

EMERALD-ID can be operated sequentially or in parallel, depending on the size of the ligand library. When operating sequentially, the cryoEM map needs to be loaded once and all ligand molecules will be docked in a single job of Rosetta. For large ligand libraries, separate EMERALD runs for each molecule can happen in parallel, and an external python script evaluates and rank all ligands once docking is complete. The runtime for individual EMERALD runs varies based on the size of the ligand, the number of rotatable side chains in the pocket, and the size of the density map. The median time for an ATP molecule is 33 minutes, but large phospholipids can take over two hours to dock. Examples on how to run in both modes are included in the file repository described in the Code Availability statement.

Ligand identification of common ligands with Phenix

EM maps for entries in the common ligand dataset were converted to structure factors using phenix.map_to_structure_factors. Ideal CIF files for each ligand in the common ligand library were downloaded from the PDB. The structure factors, an input model without the ligand, a directory containing the CIF files, and a search center of the center of mass of the deposited ligand model were provided to phenix.ligand_identification (v. 1.21.1-5286). The rankings of the deposited identity from Phenix identification were compared to EMERALD-ID.

Comparison of EMERALD-ID with the MinkLoc3Dv2 model

Labels of ligand classes used in the MinkLoc3Dv2 model were gathered from https://checkmyblob.bioreproducibility.org/server/ligands/ and were filtered to only include those that were labeled as "non-polymers" in the Chemical Component Dictionary,[24] had 5 or more heavy atoms, and did not include an ion. This created library of 103 ligand identities that could be used for EMERALD-ID. To ensure that the cases we tested had not been seen by the MinkLoc3Dv2 model during training, we searched the EMDB for entries deposited in 2024 that contained one of the 103 identities. Entries were filtered using the same criteria in the "determination of common ligand library" section, providing 429 cases.

The MinkLoc3Dv2 model takes an excised density region from a difference map as input. For each EMDB entry, the associated PDB file was stripped of all HETATM lines. The resulting structure and the full map deposited for the EMDB entry were provided to phenix.real_space_diff_map to create difference maps. Density blobs for the first instance of a ligand in the structure were extracted using the cut_ligands program provided with the MinkLoc3Dv2 model (https://github.com/dabrze/cryo-em-ligand-cutter/tree/6032b5701cad7a4db86f780b91c2078907e36e42). Not all blobs were suitable for extraction for the MinkLoc3Dv2 model, leaving 291 entries to test. Extracted blobs were evaluated using the MinkLoc3Dv2 Ligand Classification API using the rescale_cryoem option. All cases were evaluated in EMERALD-ID using the same docking settings and scoring explained above.

Unassigned density finder methodology and filtering

The EMDB ligand-bound entries within 2-6 Å global resolution described above along with 3-4 Å maps containing structures without bound ligands were searched for unassigned regions ("blobs") of the map that could possibly belong to a ligand. To discover these regions, we developed an unassigned density finding tool. The tool created a mask of the receptor and calculated the mean and standard deviation of all voxel values within the mask. Z-scores for all voxel values outside the mask were calculated using these values. Voxels with a Z-score greater than 0.5 were labeled as peaks and neighboring peak voxels were grouped together to form blobs. Each blob was scored by its number of voxels and fraction of surface voxels that are within 4 Å of the receptor. Blobs were filtered to only keep those with more than 70 voxels, more than 90% of the blob surface interacting with the receptor, and further than 5 Å from a cut or terminus in the protein structure. All blobs passing the filters were screened with the common ligand library with the binding pocket centered on the found blob. To limit the number of cases to visually inspect, top-ranked ligand identities with a Z-score greater than -0.5 were further analyzed for possible ligand identification.

Endogenous ligand library screening

We obtained 3030 SMILES strings for all detected and quantified metabolites with an endogenous origin from the Human Metabolome Database.[36] After processing as described above, we had 2950 ligand identities to use for docking. Using the RCSB REST API,[75] we searched for rare occurring identities that had one to three cryoEM solved structures containing the respective ligand's SMILES string. Entries were filtered to those that had a human source organism, a resolution worse than 3.3 Å, and more than 5 heavy atoms in the ligand. This left us with 14 entries, which were then manually pruned to 7 after removing entries with multiple ligands in the binding pocket and large lipids with inconclusive support in the EM map. The 2950 endogenous ligand identities were docked for each of the 7 entries and ranks were determined. A varying amount of ligands for each case did not complete docking because of reaching a time limit on computing resources. Ligand similarity among the endogenous library was calculated by Tanimoto similarity coefficient of small molecule fingerprints with RDkit (release 2024.03.4).[76]

Fragment screening preparation

Both examples from the fragment cocktail experiments for pyruvate kinase 2 from Saur et al.[49] were used for fragment screening. Fragment parameters were created from SMILES strings as described above and a library of the respective cocktails were provided for identification for EMERALD-ID. For examples to use with simulated data, fragment bound crystal structures were taken from Congreve et al.[77] The Cambridge[50] and York[51] 3D libraries provided 137 and 106 fragments, respectively, for the simulated data fragment screening. We chose these libraries because their SMILES strings were publicly available and the fragment sizes in the library were similar to the fragments in the crystal structures.

The sim_cryo tool[78] in Rosetta was used to simulate cryoEM maps for the fragment bound crystal structures. Briefly, sim_cryo creates 2D projections of protein structures for map reconstruction rather than attempting to directly simulate the 3D map. The input structure is randomly rotated for a selected number of rotations, and projection images across each XYZ plane are recorded for each rotation. Gaussian noise is applied to each image, and pixels in the projections, which correspond to atoms in the structure, are randomly perturbed to simulate atom heterogeneity.

We simulated cryoEM maps for the crystal structures to a resolution around 3.5 Å by using a Gaussian noise multiplier of 0.6, a pixel size of 1, and an atom perturbation factor of the atom's B-factor divided by 120 to produce an image stack of 45000 projections. The image stacks of the perturbed projections were passed into cryoSPARC (v.4.4)[79] and 2D class averages were created. 3D maps were created from 2D classes with ab initio reconstruction and then a homogeneous refinement. Two cases, 1FV9 and 2JJC, could not produce realistic cryoEM maps because of their small size, but all other structures produced simulated maps with realistic low-resolution ligand binding sites (Figures S6A–S6G). To further show the quality of simulated data with sim_cryo, we simulated EM data of cryoEM-solved structures using the same simulation protocol described above. We found that map correlations for ligand models and their binding pockets were similar for the real EM and simulated EM data (Figures S6J–S6L).

Ligand and data visualization

Figures of ligand-bound models and their EM maps were created using UCSF Chimera (v. 1.17.3).[80] Maps shown in figures were the whole maps downloaded directly from each respective EMDB entry. Plotting of data was performed using the ggplot2 package (v. 3.4.3) in R.[81]

QUANTIFICATION AND STATISTICAL ANALYSIS

Linear regression model

Linear regression models were created using the lm() function in R (v. 4.3.1) to study the relationship between ligand and map features to the estimated binding affinity and ligand density correlation. Entries of cryoEM-solved ligand-bound structures were gathered (see "Linear regression model calculation" in method details above), and entry features were used as predictors for the regression models of the estimated binding affinity (n = 558 entries) and density correlation transformed by a hyperbolic tangent function (n = 555 entries). The significance of the model coefficients were evaluated by a *t*-test, and coefficients with *p*-values less than 0.05 were included in the model. A full summary of the linear regressions for the estimated binding affinity and transformed density correlation can be found in Tables S2 and S3, respectively. To examine the assumptions of the linear models, diagnostic plots for the model were created (Figures S7A–S7H), including linearity of the residuals (Figures S7A and S7E), the quantile-quantile plots of the residuals (Figures S7B and S7F), the variance of the residuals (Figures S7C and S7G), and the influence of individual entries (Figures S7D and S7H).

Structure, Volume *33*

Supplemental Information

Automated identification of small molecules in cryoelectron microscopy data with density- and energy-guided evaluation

Andrew Muenks, Daniel P. Farrell, Guangfeng Zhou, and Frank DiMaio

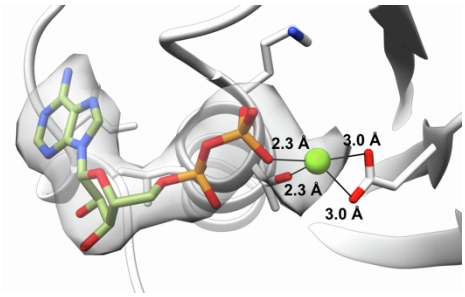


Figure S1. 8dar fitted with a magnesium ion, Related to Figure 3. Docked pose of ADP from EMERALD-ID modeled with a magnesium ion. Possible coordinate bonds and distances to nearby coordinating atoms are labeled.

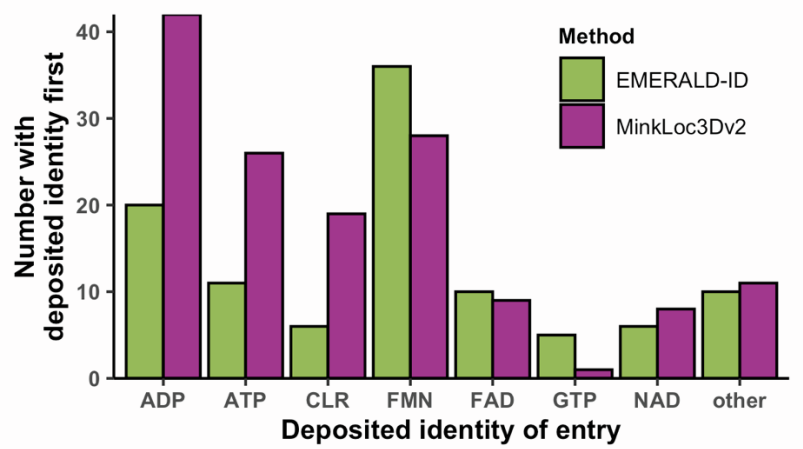


Figure S2. Comparison of identification success by the deposited identity, Related to Figure 2. The count of maps ranking the deposited identity first by EMERALD-ID and MinkLoc3Dv2 depending on the identity of the deposited ligand. MinkLoc3Dv2 outperforms EMERALD-ID in identifying ADP, ATP, and cholesterol (CLR).

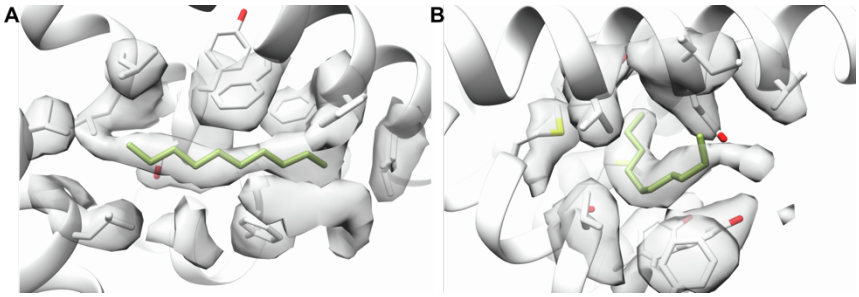
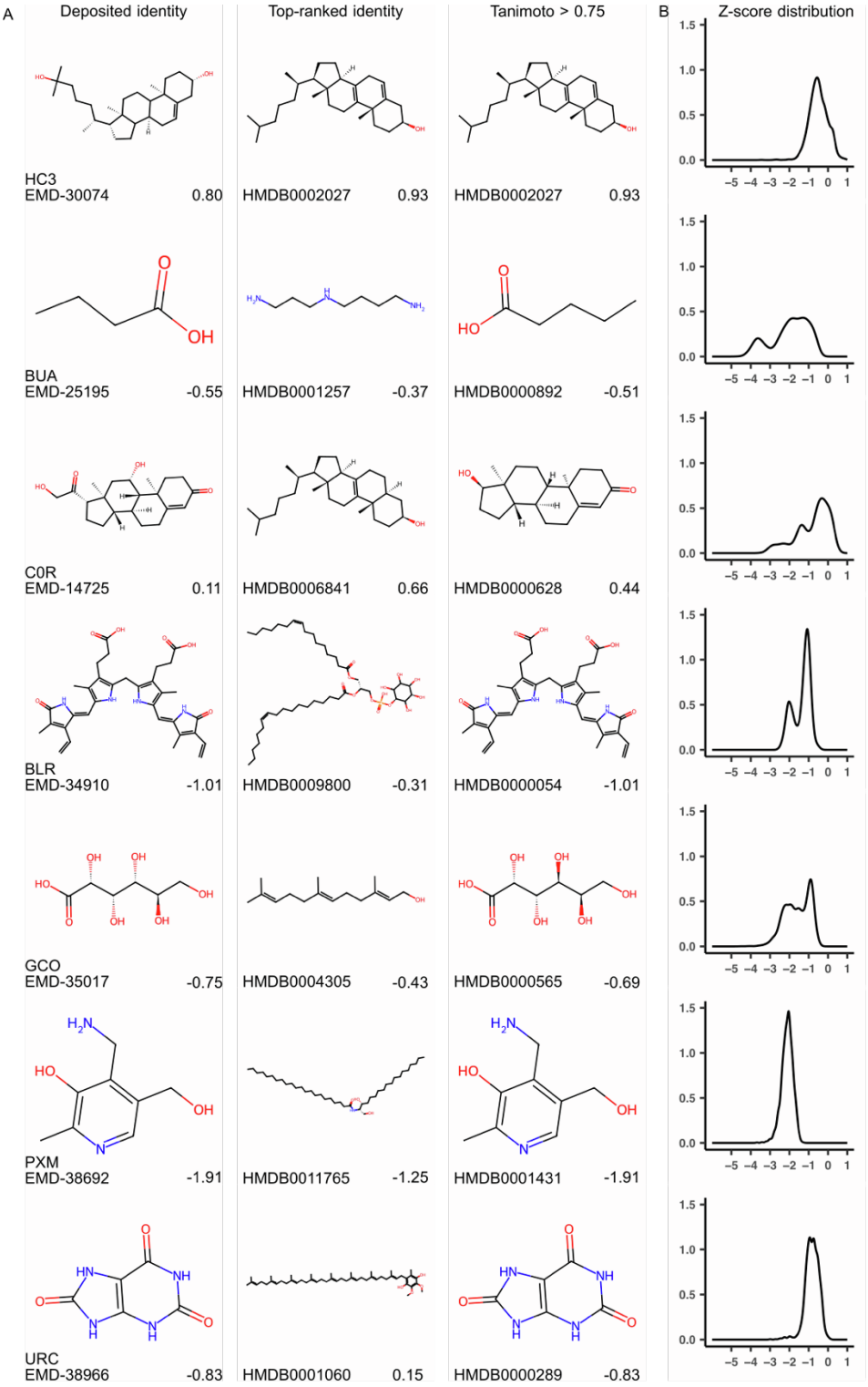


Figure S3. Lipids detected in unmodeled density regions, Related to Figure 4. (A) Ten carbon chain in transmembrane region of TRPV2 (EMDB: 33774, PDB: 7YEP). (B) Ten carbon chain in transmembrane region of TRPV1 (EMDB: 23128, PDB: 7L2H). Map for (A) is zoned 5.0 Å around ligand at a 0.0148 contour level (σ = 0.4). Map for (B) is zoned 5.0 Å around ligand at a 0.0055 contour level (σ = 0.3).


A
Deposited identity
Top-ranked identity
Tanimoto > 0.75
B
Z-score distribution
HC3
EMD-30074 0.80
HMDB0002027 0.93
HMDB0002027 0.93
BUA
EMD-25195 -0.55
HMDB0001257 -0.37
HMDB0000892 -0.51
C0R
EMD-14725 0.11
HMDB0006841 0.66
HMDB0000628 0.44
BLR
EMD-34910 -1.01
HMDB0009800 -0.31
HMDB0000054 -1.01
GCO
EMD-35017 -0.75
HMDB0004305 -0.43
HMDB0000565 -0.69
PXM
EMD-38692 -1.91
HMDB0011765 -1.25
HMDB0001431 -1.91
URC
EMD-38966 -0.83
HMDB0001060 0.15
HMDB0000289 -0.83
1.5
1.0
0.5
0.0
−5 −4 −3 −2 −1 0 1

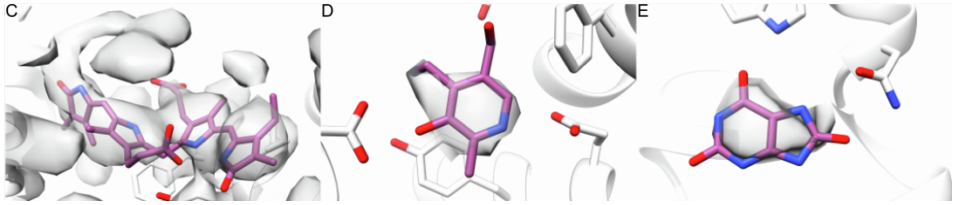

C
D
E

Figure S4. Structures, Z-score distributions, and EM maps for cases screened with an endogenous ligand library, Related to Table 1. (A) 2D drawings of molecules from endogenous ligand screen. The deposited identity (left), top-ranked identity (center), and highest-ranking identity with a Tanimoto score > 0.75 to the deposited identity (right) for each entry in Table 1 are shown. Accession numbers for the molecule in the Human Metabolome Database and the Z-score for the identity are shown below each drawing. (B) Kernel density plot of Z-scores of all identities for each case. (C-E) Deposited ligand models for endogenous ligand screens without a similar ligand in the top 10 percent. (C) Bilirubin bound to OATP1B1 (EMDB: 34910, PDB: 8HNC, Chain: A, Residue: 701). (D) Pyridoxamine bound to SLC19A3 (EMDB: 38692, PDB: 8XV5, Chain: A, Residue: 501). (E) Urate bound to GLUT9 (EMDB: 38966, PDB: 8Y65, Chain: A, Residue: 601). Map for (C) is zoned 6.0 Å around ligand at a 0.467 contour level (σ = 1.0). Map for (D) is zoned 2.0 Å around ligand at a 0.0898 contour level (σ = -0.15). Map for (E) is zoned 2.0 Å around ligand at a 1.13 contour level (σ = 0.7).

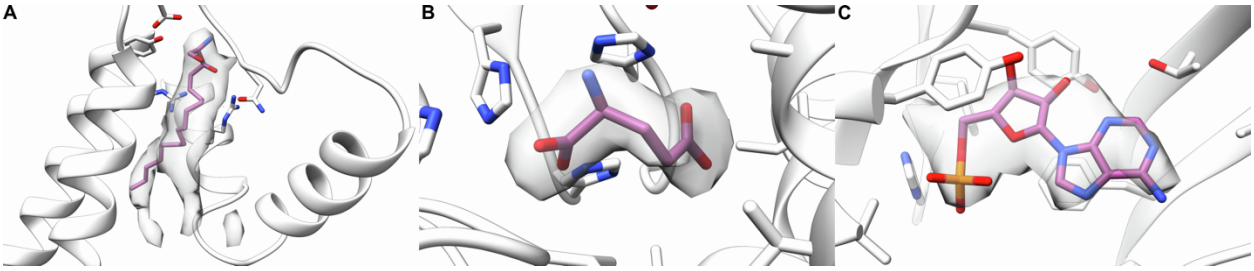


Figure S5. Top common ligand identity for detected density, Related to Figure 5. (A) Detected sphingosine lipid in the Piezo 1 ion channel (EMDB: 7128, PDB: 6BPZ). (B) Detected glutamate molecule in ADH3 from S. acidaminiphila (EMDB: 35452, PDB: 8IHQ). (C) Detected AMP ligand in a $GABA_A$ receptor (EMDB: 40462, PDB: 8SGO). Map for (A) is zoned 2.0 Å around ligand at a 0.0202 contour level ($\sigma = 0.35$). Map for (B) is zoned 2.5 Å around ligand at a 0.26 contour level ($\sigma = 0.6$). Map for (C) is zoned 2.0 Å around ligand at a 0.0369 contour level ($\sigma = 1.1$).

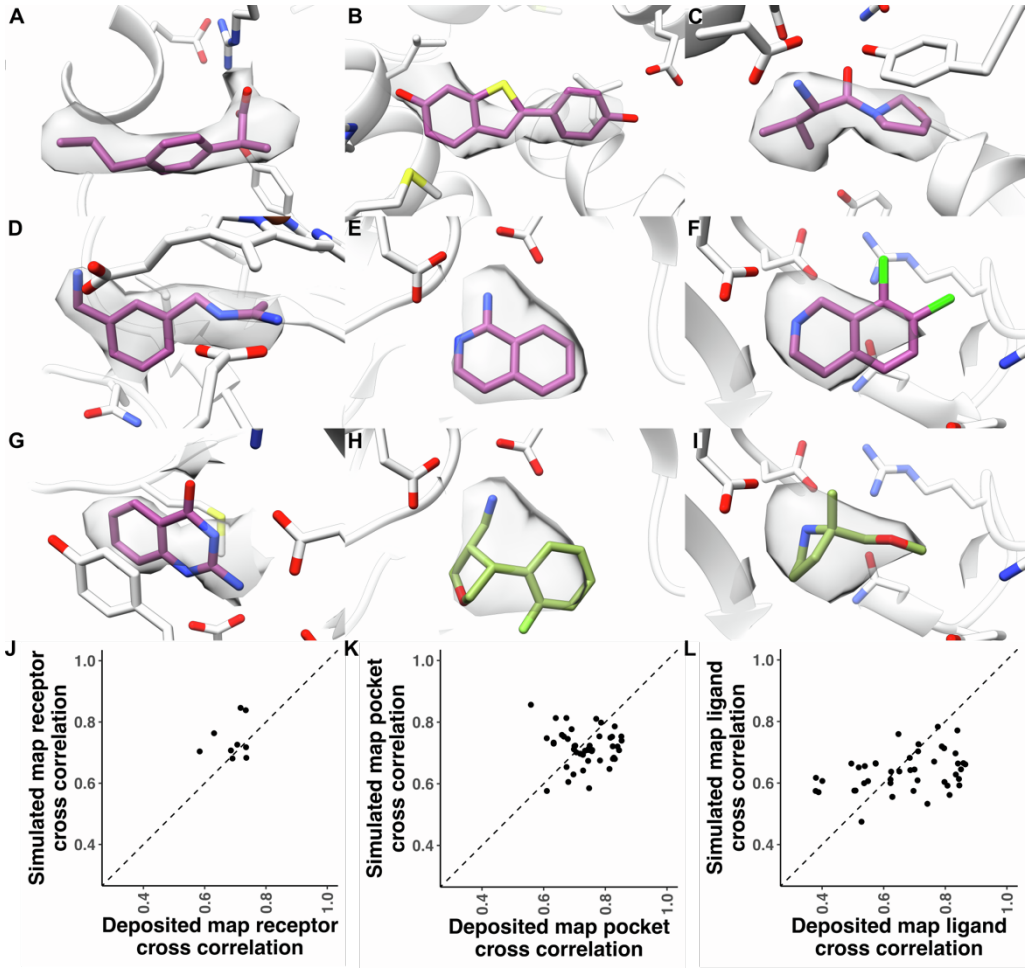


Figure S6. Simulated EM map information for drug fragments, Related to Table 2. (A-G) Deposited ligands with their simulated EM maps. (A) Ibuprofen bound to COX-1 (PDB: 1EQG, Chain: A, Residue: 701). (B) Raloxifene core bound to estrogen receptor alpha (PDB: 1GWQ, Chain: A, Residue: 600). (C) Fragment inhibitor bound to dipeptidyl peptidase IV/CD26 (PDB: 1N1M, Chain: A, Residue: 954). (D) W1400 inhibitor bound to nitric oxide synthase (PDB: 1QWC, Chain: A, Residue: 902). (E) 1-amino-isoquinoline bound to beta-secretase (PDB: 2OHK, Chain: A, Residue: 390). (F) Fragment inhibitor bound to phenylethanolamine N-methyltransferase (PDB: 1YZ3, Chain: A, Residue: 3001). (G) Fragment inhibitor bound to tRNA-guanine transglycosylase (PDB: 1S39, Chain: A, Residue: 900). (H, I) Top ligand identities for failed fragment screening cases. (H) Top ranked fragment for a simulated EM map of beta-secretase (PDB: 2OHK, Chain: A, Residue: 390). (I) Top ranked fragment for a simulated EM map of phenylethanolamine N-methyltransferase (PDB: 1YZ3, Chain: A, Residue: 3001). Map for (A) is zoned 2.0 Å around ligand at a 2.90 contour level (σ = 0.8). Map for (B) is zoned 2.0 Å around ligand at a 2.63 contour level (σ = 0.85). Map for (C) is zoned 2.0 Å around ligand at a 3.12 contour level (σ = 1.0). Map for (D) is zoned 2.0 Å around ligand at a 2.79 contour level (σ = 1.0). Map for (E, H) is zoned 2.0 Å around ligand at a 2.68 contour level (σ = 0.8). Map for (F, I) is zoned 2.0 Å around ligand at a 2.45 contour level (σ = 0.7). Map for (G) is zoned 2.0 Å around ligand at a 2.52 contour level (σ = 0.5). (J-L) Comparison of select deposited EM maps to their simulated data. (J) Comparison of cross correlation of the receptor to deposited and simulated density maps for 9 EMDB entries. (K) Comparison of cross correlation of the ligand binding pocket to deposited and simulated density maps. Each point corresponds to a small molecule residue of the 9 maps. Pocket is defined as 10 Å around the center of mass of the ligand. (L) Comparison of cross correlation of the ligand model to deposited and simulated density maps for 9 EMDB entries. Each point corresponds to a small molecule residue of the 9 maps.

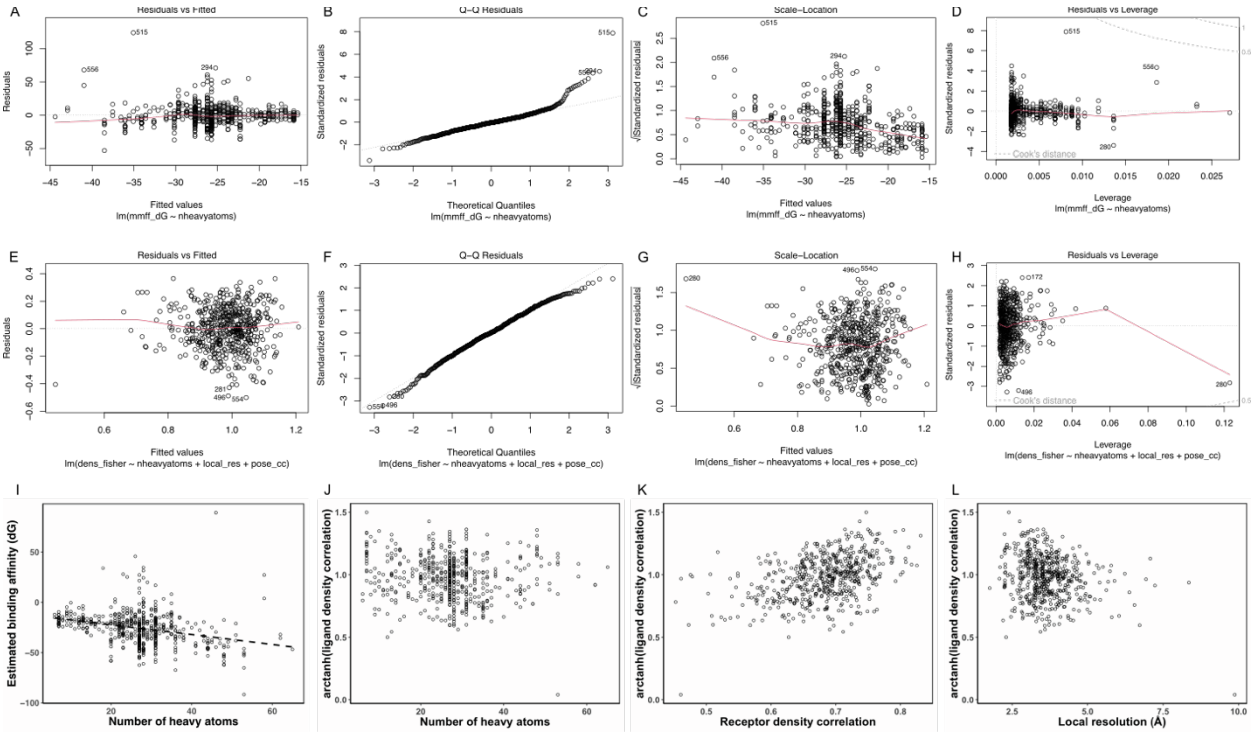


Figure S7. Diagnostic plots for linear regression model, Related to STAR Methods. (A-D) Diagnostic plots provided by R's linear regression for the estimated binding affinity linear regression. mmff_dG is the estimated binding affinity in Rosetta. nheavyatoms is the number of heavy atoms in a molecule. (E-H) Diagnostic plots provided by R's linear regression for the ligand density correlation linear regression. lig_dens is the ligand density correlation. nheavyatoms is the number of heavy atoms in a molecule. pose_cc is the density correlation of the receptor to the EM map. local_res is the local resolution of the EM map 10 Å around the ligand. (I-L) Scatterplots of predictor and response variable values used for the linear regression.

Table S1. List of identities used for common ligand library, Related to Figure 2.

CCD ID	Name
1N7	chapso
8Q1	s-[2-(n-[(2r)-2-hydroxy-3,3-dimethyl-4-(phosphonooxy)butanoyl]-beta-alanylamino)ethyl] dodecanethioate
9Z9	(3beta,14beta,17beta,25r)-3-[4-methoxy-3-(methoxymethyl)butoxy]spirost-5-en
ABU	gamma-amino-butanoic acid
ADP	adenosine-5'-diphosphate
AMP	adenosine monophosphate
ATP	adenosine-5'-triphosphate
BCR	beta-carotene
CLR	cholesterol
D10	decane
D12	dodecane
FAD	flavin-adenine dinucleotide
FMN	flavin mononucleotide
GDP	guanosine-5'-diphosphate
GLU	glutamic acid
GTP	guanosine-5'-triphosphate
IHP	inositol hexakisphosphate
LMT	dodecyl-beta-d-maltoside
LUT	(3r,3'r,6s)-4,5-didehydro-5,6-dihydro-beta,beta-carotene-3,3'-diol
NAD	nicotinamide-adenine-dinucleotide
NAI	1,4-dihydronicotinamide adenine dinucleotide
NAP	nadp nicotinamide-adenine-dinucleotide phosphate
NDP	nadph dihydro-nicotinamide-adenine-dinucleotide phosphate
NEX	(1r,3r)-6-(3e,5e,7e,9e,11e,13e,15e,17e)-18-[(1s,4r,6r)-4-hydroxy-2,2,6-trimethyl-7-oxabicyclo[4.1.0]hept-1-yl]-3,7,12,16-tetramethyloctadeca-1,3,5,7,9,11,13,15,17-nonaenylidene-1,5,5-trimethylcyclohexane-1,3-diol
PLM	palmitic acid
PQN	phylloquinone
SAM	s-adenosylmethionine
SPH	sphingosine
XAT	(3s,5r,6s,3's,5'r,6's)-5,6,5',6'-diepoxy-5,6,5',6'-tetrahydro-beta,beta-carotene-3,3'-diol
Y01	cholesterol hemisuccinate

Table S2. Summary statistics for estimated binding affinity linear regression, Related to STAR Methods.

MODEL INFO:

Observations: 558
Dependent Variable: dG
Type: OLS linear regression

MODEL FIT:

F(1,556) = 55.3567, p = 0.0000
R^2 = 0.0905
Adj. R^2 = 0.0889
Residual standard error = 15.8

Standard errors: OLS	Est.	S.E.	t val.	p
(Intercept)	-12.4443	1.9060	-6.5289	0.0000
nheavyatoms	-0.4918	0.0661	-7.4402	0.0000

Table S3. Summary statistics for ligand density correlation transformed with arctanh linear regression, Related to STAR Methods.

MODEL INFO:

Observations: 555 (3 missing obs. deleted)
Dependent Variable: arctanh(ligand density)
Type: OLS linear regression

MODEL FIT:

F(3,551) = 54.99, p = 0.0000
R^2 = 0.2304
Adj. R^2 = 0.2262
Residual standard error = 0.1531

Standard errors: OLS	Est.	S.E.	t val.	p
(Intercept)	0.3428	0.0825	4.156	0.0000
nheavyatoms	-0.0015	0.0007	-2.223	0.0266
pose_cc	1.1935	0.1073	11.119	0.0000
local_res	-0.0372	0.0077	-4.836	0.0000